Sneha Kulkarni

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Professional Summary

Statistician turned data scientist with five years of experience in healthcare and pharma analytics. Specialises in survival analysis, causal inference and rigorous study design, working in both R and Python. Emphasis on interpretability, validation documentation and clear communication to clinical and commercial stakeholders.

Technical Skills

Programming: R (tidyverse, survival, lme4), Python (Pandas, statsmodels, scikit-learn), SQL

Statistics: Survival analysis (Kaplan-Meier, Cox models), mixed-effects models, propensity score matching, power analysis

Experimentation: Observational study design, difference-in-differences, A/B test analysis

Machine Learning: Regularised regression, random forest, gradient boosting, model calibration

Visualization and Reporting: Tableau, ggplot2, R Markdown, Matplotlib

Data Handling: EDA, missing data (multiple imputation), data validation

Work Experience

Data Scientist - Konkan HealthMetrics Pvt Ltd
Mumbai, Maharashtra | *Apr 2024 - Present*

- Lead analyses on patient adherence for a chronic-care programme, using Cox models with time-varying covariates to quantify the effect of reminder frequency on discontinuation.
- Used propensity score matching on about 48,000 patients to estimate programme impact; the matched analysis showed an 11% relative improvement in 6-month adherence.
- Produce validation reports for every model, covering calibration, subgroup performance and sensitivity analysis, reviewed by a medical advisory group.
- Mentor two junior analysts on study design and reproducible R workflows.

Associate Data Scientist - Konkan HealthMetrics Pvt Ltd
Mumbai, Maharashtra | *Aug 2021 - Mar 2024*

- Built a hospital readmission risk model (regularised logistic regression with calibration) that reached ROC-AUC 0.74 while remaining explainable to clinicians.
- Applied multiple imputation to lab data with 30% missingness and compared results against complete-case analysis.
- Created Tableau dashboards on programme enrolment and outcomes for commercial teams.

Projects

Sales Force Effectiveness Study

- Problem: Determine whether additional field visits improved prescription uptake.
- Approach: Applied difference-in-differences with matched control territories and tested parallel-trends assumptions.
- Technologies: R, Python, statsmodels
- Results: Estimated a 4.2% uplift attributable to extra visits, informing the next territory plan.

Reproducible Analysis Template

- Problem: Analyses were difficult to reproduce across team members.
- Approach: Created an R Markdown project template with environment locking, data checks and automatic report generation.
- Technologies: R, renv, Git
- Results: Adopted across the analytics team for new studies.

Education

M.Sc., Statistics, Konkan School of Engineering and Sciences, Mumbai (2019 - 2021) - First class with distinction

B.Sc., Statistics, Western Ghats College of Science, Mumbai (2016 - 2019)

Certifications

- Statistical Learning (Stanford Online, 2022)
- Tableau Certified Data Analyst (2023)

Achievements

- Led an analysis that shaped a programme redesign for patients across four states.
- Paper summary on propensity score methods presented at an internal journal club, 2024.